

Q1 - 25 July - Shift 1

The number of solutions of $|\cos x| = \sin x$, such that $-4\pi \leq x \leq 4\pi$ is :

- (A) 4 (B) 6 (C) 8 (D) 12

Space for your notes:

Q2 - 26 July - Shift 1

Let

$S = \{\theta \in [0, 2\pi] : 8^{2\sin^2 \theta} + 8^{2\cos^2 \theta} = 16\}$. Then

$n(S) + \sum_{\theta \in S} \left(\sec\left(\frac{\pi}{4} + 2\theta\right) \operatorname{cosec}\left(\frac{\pi}{4} + 2\theta\right) \right)$ is

equal to :

- (A) 0 (B) -2
(C) -4 (D) 12

Space for your notes:

Q3 - 26 July - Shift 2

If the sum of solutions of the system of equations

$2\sin^2 \theta - \cos 2\theta = 0$ and $2\cos^2 \theta + 3\sin \theta = 0$ in the

interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

Space for your notes:

Q4 - 29 July - Shift 1

Let $S = \{\theta \in (0, 2\pi) : 7\cos^2 \theta - 3\sin^2 \theta - 2\cos^2 2\theta = 2\}$. Then, the sum of roots of all the equations $x^2 - 2(\tan^2 \theta + \cot^2 \theta)x + 6\sin^2 \theta = 0$ $\theta \in S$, is _____.

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Answer Key

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Q1 (C)

Q2 (C)

Q3 (3)

Q4 (16)

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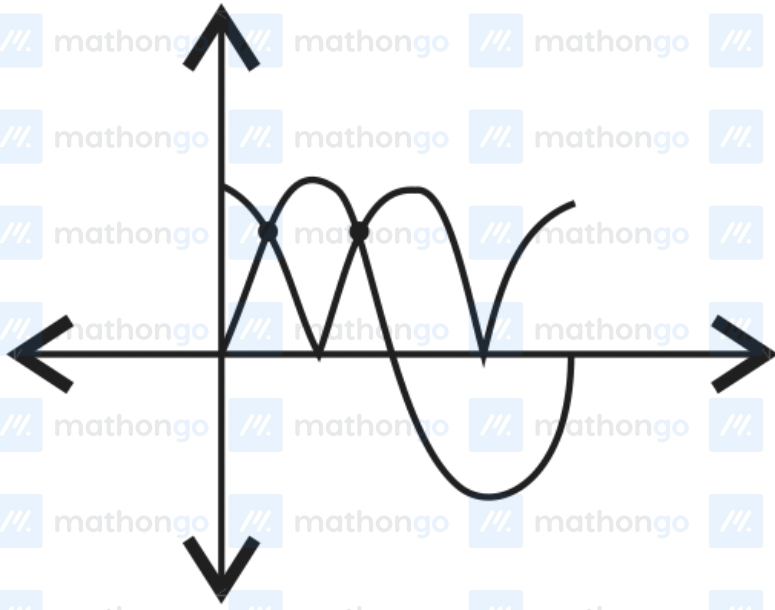
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Q1 (C)



2 solutions in $(0, 2\pi)$

So, 8 solutions in $[-4\pi, 4\pi]$

Q2 (C)

$$8^{2\sin^2\theta} + 8^{2-2\sin^2\theta} = 16$$

$$y + \frac{64}{y} = 16$$

$$\Rightarrow y = 8$$

$$\Rightarrow \sin^2\theta = \frac{1}{2}$$

$$n(S) + \sum_{\theta \in S} \frac{1}{\cos(\pi/4 + 2\theta)\sin(\pi/4 + 2\theta)}$$

$$= 4 + (-2) \times 4 = -4$$

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Q3 (3)

$$2\sin^2 \theta - \cos 2\theta = 0$$

$$2\sin^2 \theta - (1 - 2\sin^2 \theta) = 0$$

$$\Rightarrow \sin^2 \theta = \left(\frac{1}{2}\right)^2$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$2\cos^2 \theta + 3\sin \theta = 0$$

$$\Rightarrow 2\sin^2 \theta - 3\sin \theta - 2 = 0$$

$$\therefore \sin \theta = -\frac{1}{2}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

So, the common solution is

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

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$$\sum = \frac{\pi + 1 + \pi}{6} = 3\pi = k\pi$$

$$K = 3$$

Q4 (16)

$$7\cos^2\theta - 3\sin^2\theta - 2\cos^22\theta = 2$$

$$4\cos^2\theta + 3\cos2\theta - 2\cos^22\theta = 2$$

$$2(1 + \cos2\theta) + 3\cos2\theta - 2\cos^22\theta = 2$$

$$2\cos^22\theta - 5\cos2\theta = 0$$

$$\cos2\theta(2\cos2\theta - 5) = 0$$

$$\cos2\theta = 0$$

$$2\theta = (2n + 1)\frac{\pi}{2}$$

$$\theta = (2n + 1)\frac{\pi}{4}$$

$$S = \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$

For all four values of θ

$$x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0$$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\text{Sum of roots of all four equations} = 4 \times 4 = 16.$$